MUHAMMAD OMER

Data Science Undergraduate | Aspiring Data Analyst / ML Practitioner

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PROFESSIONAL SUMMARY

Data Science undergraduate specializing in SQL analytics, machine learning, and business intelligence. Built end-to-end analytics projects using Python, SQL, and Power BI on datasets containing over 100K records. Seeking an entry-level Data Analyst or Data Scientist role.

LEADERSHIP & VOLUNTEER EXPERIENCE

Director of Research & Innovation (Volunteer), Grow Pakistan Innovation Network *Jul 2026 – Present*

Research · Data Analysis · Innovation Leadership

Leading university partnership outreach and workshop planning to expand research culture and student engagement nationally.

TRAINING & SIMULATIONS

Data Analytics Virtual Program, DecodeLabs *May 2026 – Jun 2026*

Self-paced virtual job-simulation program

- Cleaned and analyzed real-world datasets, applying EDA and visualization techniques to surface trends across multiple data sources.
- Built visual reports (charts, summary tables) and a final analytics report following a documented, repeatable workflow from raw data to recommendation.

Data Science Virtual Program, Ai Data House *Jul 2025 – Aug 2025*

Self-paced virtual job-simulation program

- Applied core data science workflows — data cleaning, exploratory analysis, and model evaluation — to real datasets as part of a guided industry training program.
- Practiced foundational machine learning techniques (regression, classification) and built habits in version control and reproducible analysis.

PROJECTS

HR Employee Attrition SQL Analysis — Data Analyst *2025*

MySQL · Conditional Aggregation · Subqueries · CTEs · Window Functions · GitHub

- Analyzed IBM's 1,470-employee HR dataset across 10 SQL queries to identify the strongest drivers of employee attrition.
- Discovered Sales Representatives had a 39.76% attrition rate, frequent travelers left at nearly 3x the rate of non-travelers (24.91% vs 8%), and employees in their 20s left at double the rate of those in their 30s.
- Used SUM(CASE WHEN) conditional aggregation, CTE-based subqueries, and RANK() OVER (PARTITION BY) for within-department salary ranking.
- Recommended targeted retention incentives and workload adjustments for high-risk groups — young, frequently-traveling Sales Representatives — to directly reduce attrition costs.

Olist E-Commerce Analytics Platform — Data Analyst / Analytics Engineer *2025*

MySQL · Advanced SQL · Window Functions · CTEs · RFM Segmentation · Python · Pandas · Power BI · Jupyter · GitHub

- Designed a relational MySQL database from scratch (8 tables, ~100K orders), handling NULL dates, malformed rows, and missing values in raw CSVs.
- Wrote 13 SQL queries (CTEs, window functions, RFM segmentation) answering revenue trend, cohort retention, and delivery performance questions; built parallel Python EDA (cleaning, feature engineering, cohort heatmap) and a 3-page Power BI executive dashboard.
- Findings: $13.59M total revenue, SP state ~40% of revenue, repeat customers only ~3% of base, Nov 2017 revenue spiked to ~$988K (Black Friday) — full pipeline documented on GitHub with notebook, cleaned dataset, .pbix file, and README.

Fleet Intelligence Platform — ML Engineer / Operations Research *2024*

Python · Scikit-learn · SciPy (Linear Programming) · Streamlit · GitHub

- Built a logistics control-tower platform forecasting city-zone delivery demand 36 hours ahead with a Random Forest model ($R^2 = 0.94$) using spatial-temporal features like order lag and rolling-mean volume.
- Designed a prescriptive optimization layer with SciPy linear programming to reallocate idle fleet from surplus to deficit zones at minimum cost.
- Translated model and optimizer output into business KPIs — protected revenue and SLA delay reduction — deployed live via Streamlit.

Real-Time License Plate Recognition — Computer Vision / ML Engineer *2024*

Python · YOLOv8 · EasyOCR · OpenCV · Flask · Streamlit · GitHub

- Built an end-to-end AI pipeline detecting vehicles with YOLOv8, localizing plates via Haar Cascades and contour heuristics, and reading text with EasyOCR.
- Achieved ~0.6 seconds/frame on CPU — 15–20x faster than baseline — through async threads, lazy model loading, and resolution capping.
- Deployed a dark-mode web dashboard (Flask + MJPEG stream) with live webcam mode and CLI support, hosted live on Streamlit.

TECHNICAL SKILLS

Advanced: Python, SQL, Pandas, Power BI
Intermediate: Scikit-learn, Git, Tableau, MySQL, MS SQL, Jupyter, Matplotlib, Seaborn
Basic: TensorFlow, PyTorch, MongoDB, C++, Flask, VS Code

EDUCATION

BS Data Science, PAF-IAST, Haripur *Expected 2028*

FSc Computer Science, Jinnah Jam-e School & College *2024*

CERTIFICATIONS

Certified Data Scientist — PM Accelerator · Data Analytics Job Simulation — Deloitte · Additional: HackerRank SQL, Simplilearn Power BI Fundamentals